27. Reverse Proxies: TLS Termination and Headers

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Phase: Phase 10 — Load balancing and proxies
Estimated time: 1–2 days
Slug: `27-reverse-proxy`

Learning objectives

- Terminate TLS at proxy
- Pass X-Forwarded-* carefully
- Preserve client IP with real_ip
- Log correctly

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

```
# App sees proxy IP unless headers configured
curl -H 'X-Forwarded-For: 1.2.3.4' http://localhost:8080/ || true

# Test proxy locally
curl -I https://localhost --cacert lab-ca.pem
curl -I http://localhost/api
```

Trust `X-Forwarded-For` **only** from trusted proxies — otherwise clients can spoof.

Common mistakes

- Trusting XFF from the internet
- Mixed content / wrong redirect schemes
- Huge header sizes without limits

Exercises

Work in `exercises/27-reverse-proxy/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/27-reverse-proxy/` until after an honest attempt.

Self-check

1. TLS termination meaning?
2. Why not trust XFF blindly?

3. real_ip purpose?

Further reading

- https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Forwarded-For